

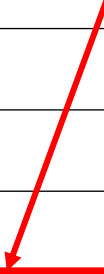
“Mechatronics lab 1 (robot: basic)”

Week 4

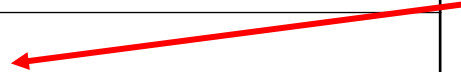
Schedule

1	Exercise (microcontroller)
2	Exercise (motors & sensors)
3	Exercise (example robot)
4	Exercise (example robot) & robot design plan
5	Exercise (example robot) & robot design plan
6	Robot design plan
7	Mid-term presentation (<u>each group</u>)
8	Robot making
9	Robot making
10	Robot making
11	Robot making
12	Robot making
13	Robot making
14	Robot making
15	Final presentation and demonstration(<u>each group</u>)

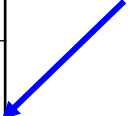
Here



Presentation about
robot design plan



Submit a final report
(each student)

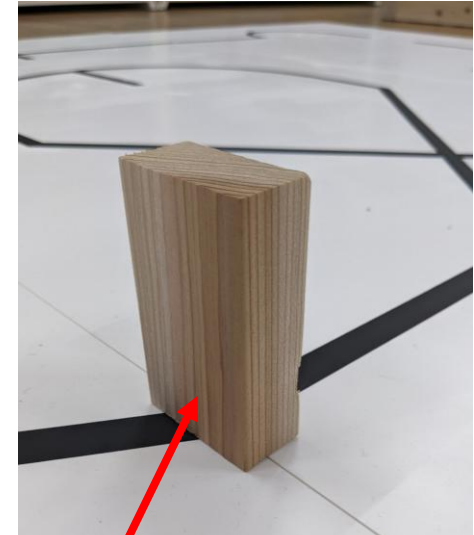


Mid-term presentation (each team)

- 5/25 (Mon), 13:00~
 - The order of presentations: Team 1, 2, 3, ..., 20
 - In English
- Presentation (7min.) + Q&A (3min.) for each team
- Contents of presentation
 - Outline of robot design plan
 - Future plan, etc.

Robot design

- Design a new robot based on your own concept (improvement of **performance**, **originality**, and so on)
- Compete with complete time of a given course (next slide)
- Two kinds of **trials** in the final demonstration
 - without obstacles
 - with obstacles (robots cannot touch obstacles)
- **Budget for components:**
 - Components prepared in advance
 - you can receive immediately (for free)
 - Components not prepared in advance
 - you order from E-portal (Budget: 10,000 yen/team)

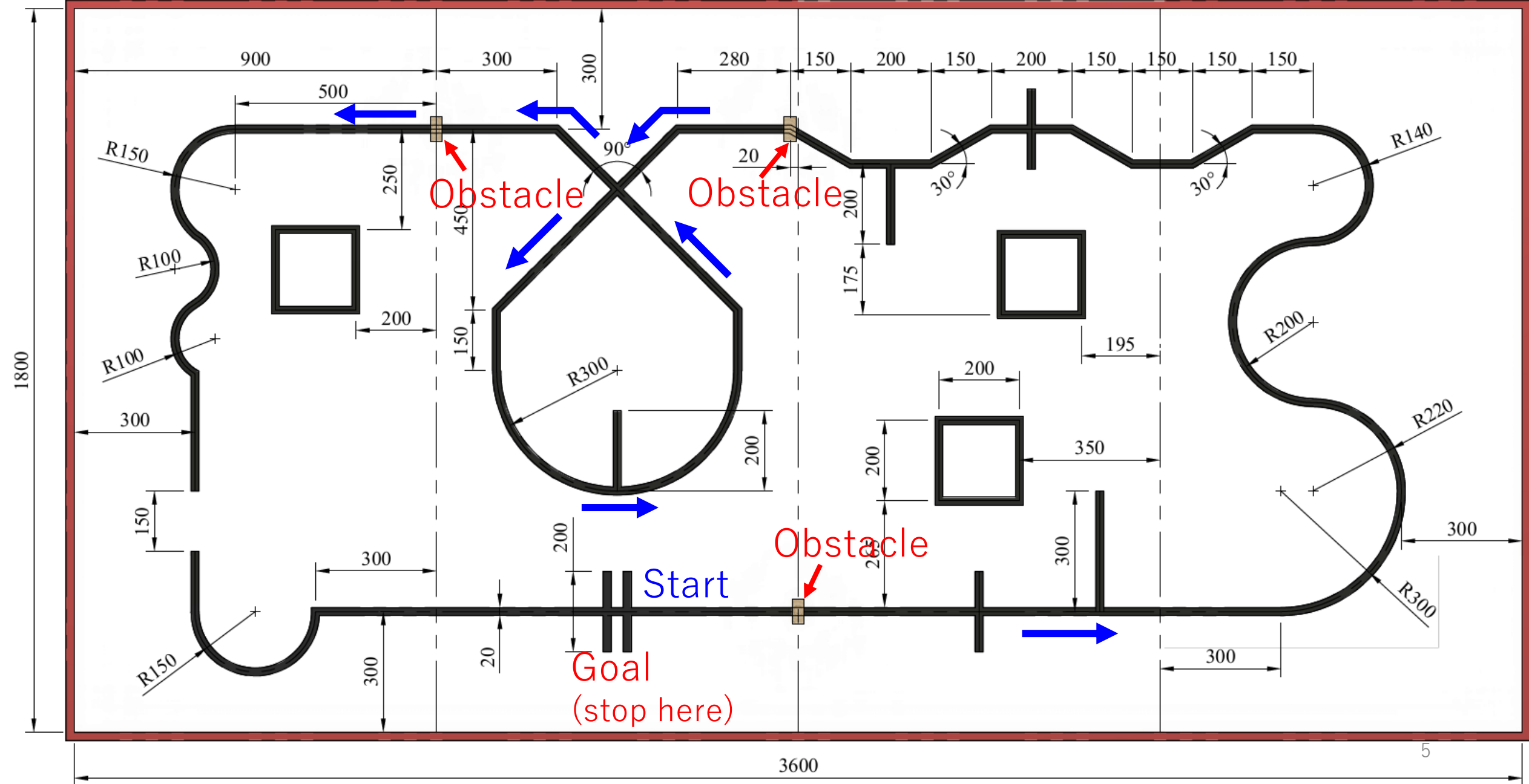


30 x 60 x 100 mm

Field

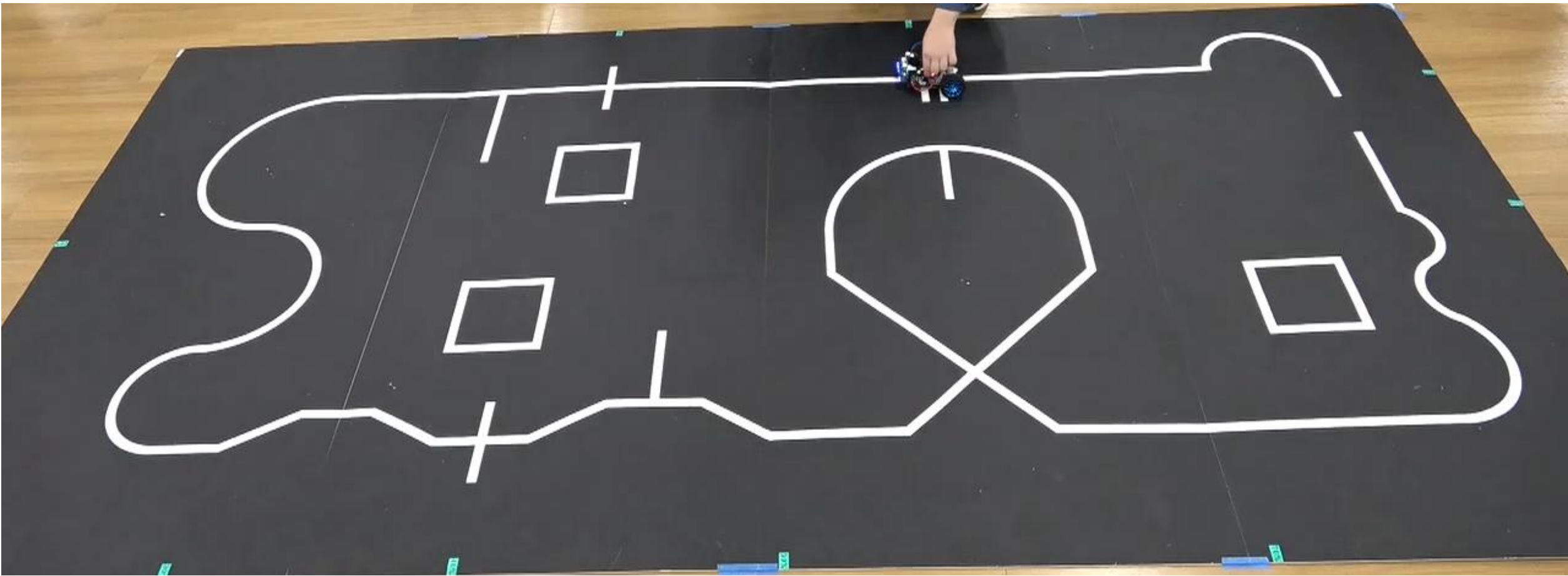
Line width: 20mm

Unit: mm



Fastest robot in 2025: 14.4 sec

➤ Trial without obstacles

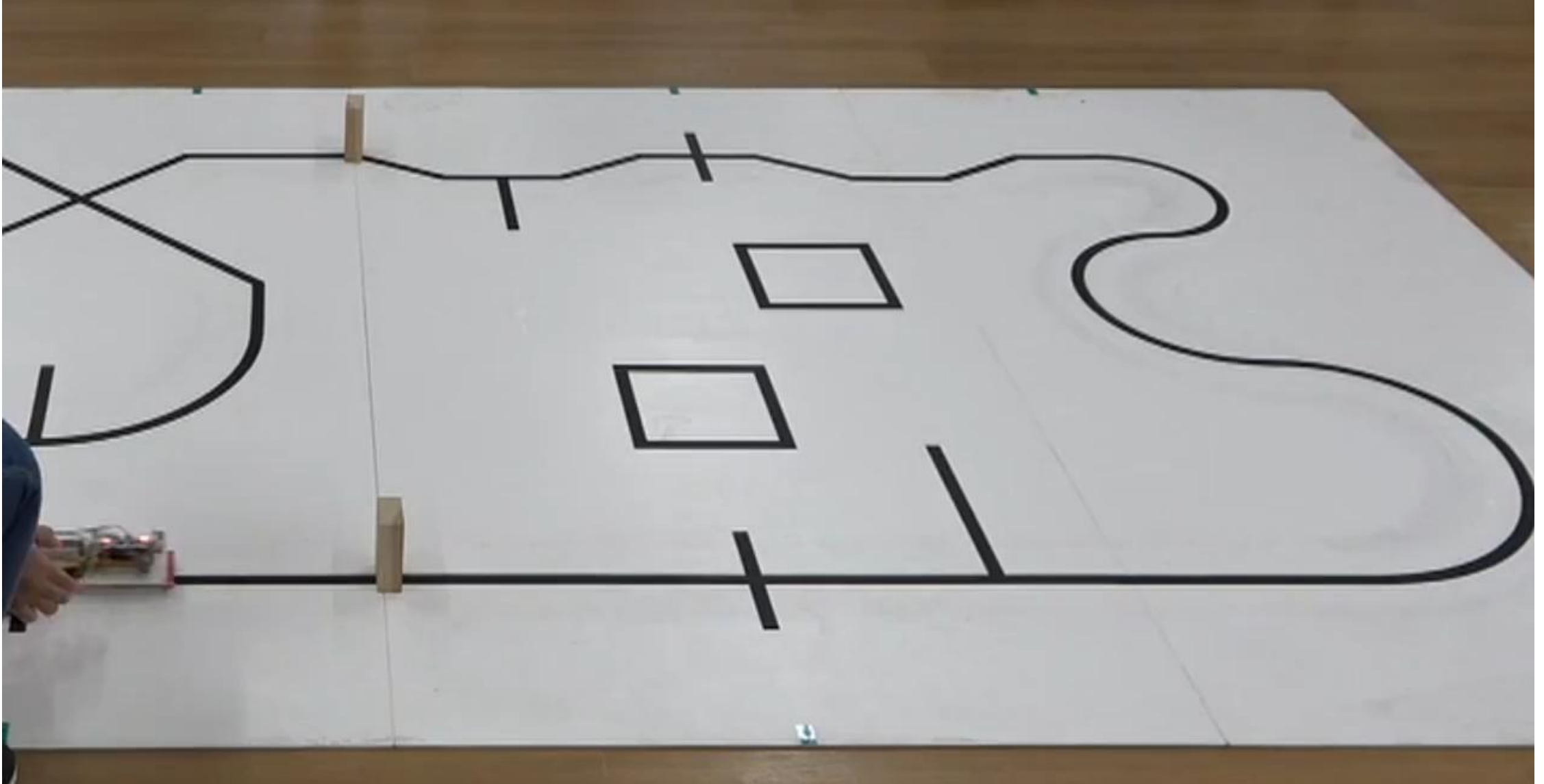


[Link to the movie](#)

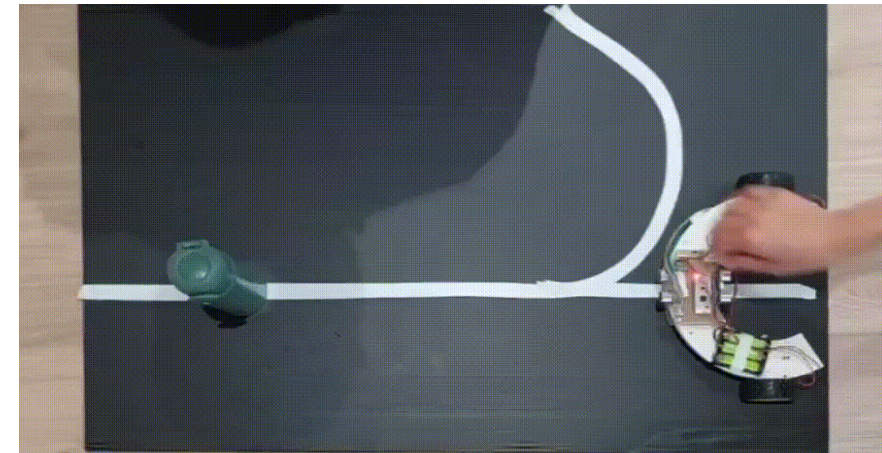
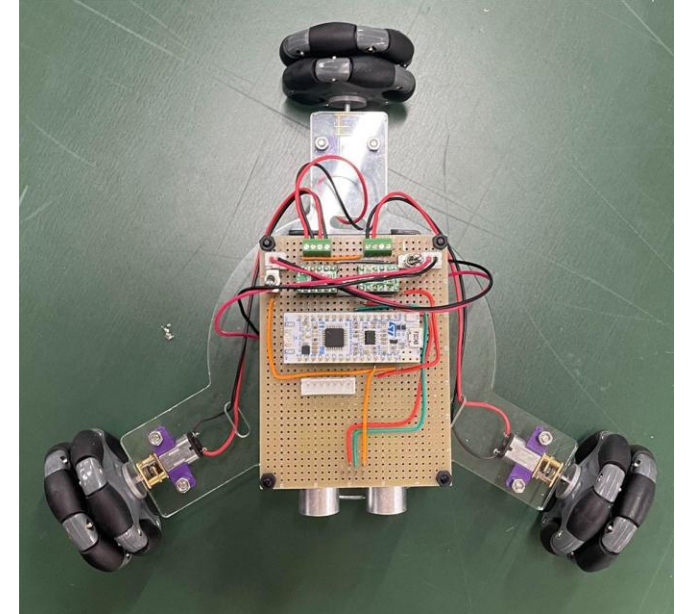
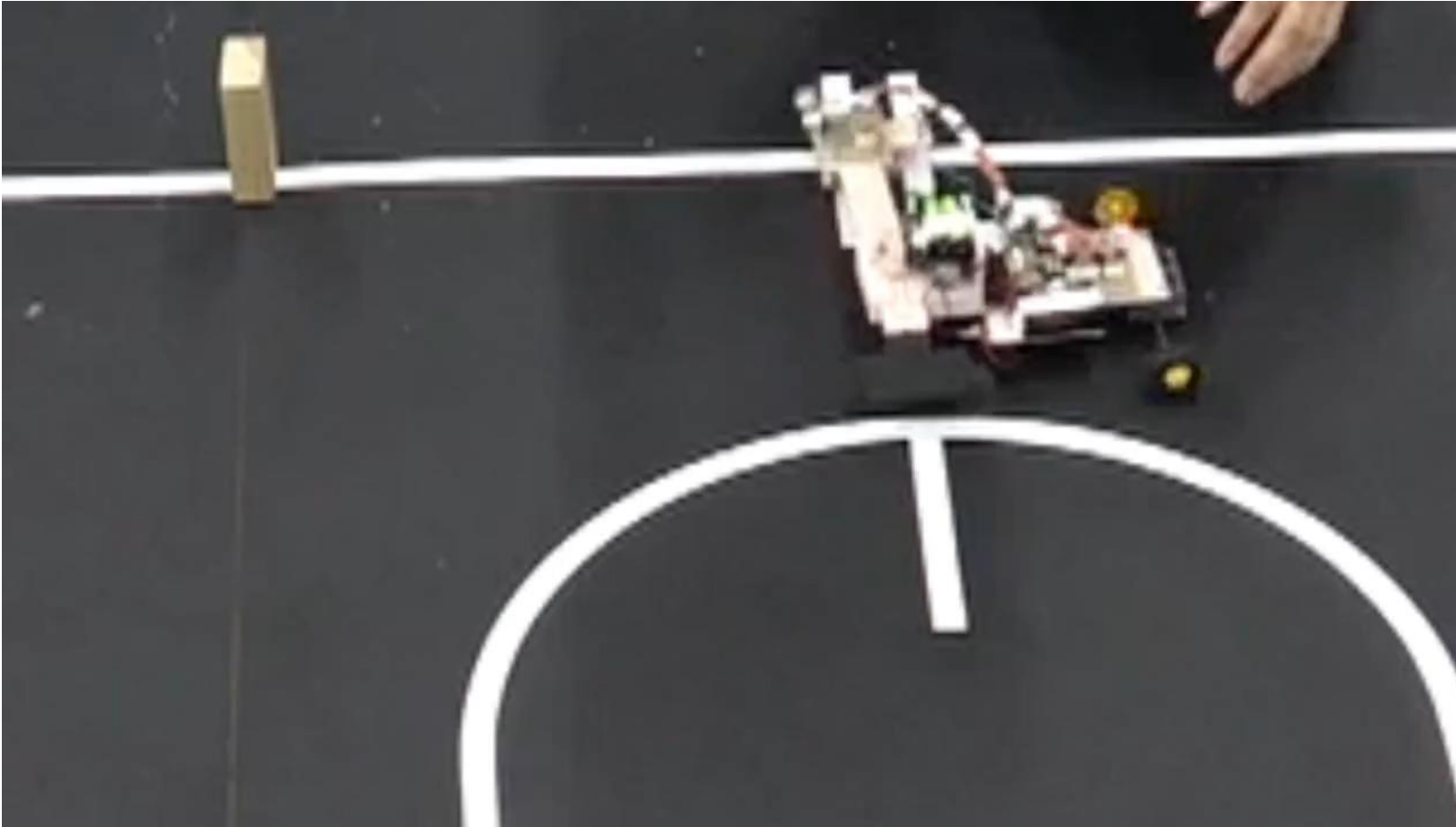
Fastest robot in 2024: 32.9 sec

➤ Trial with obstacles

[Link to the movie](#)

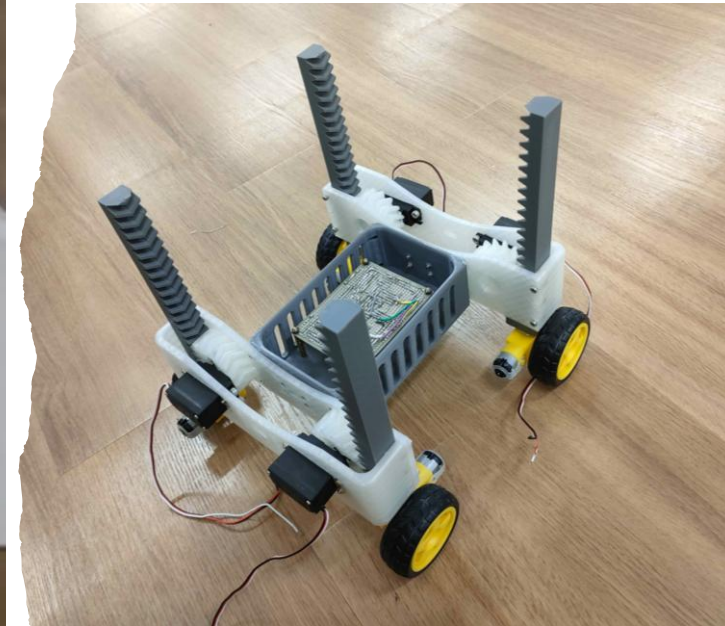
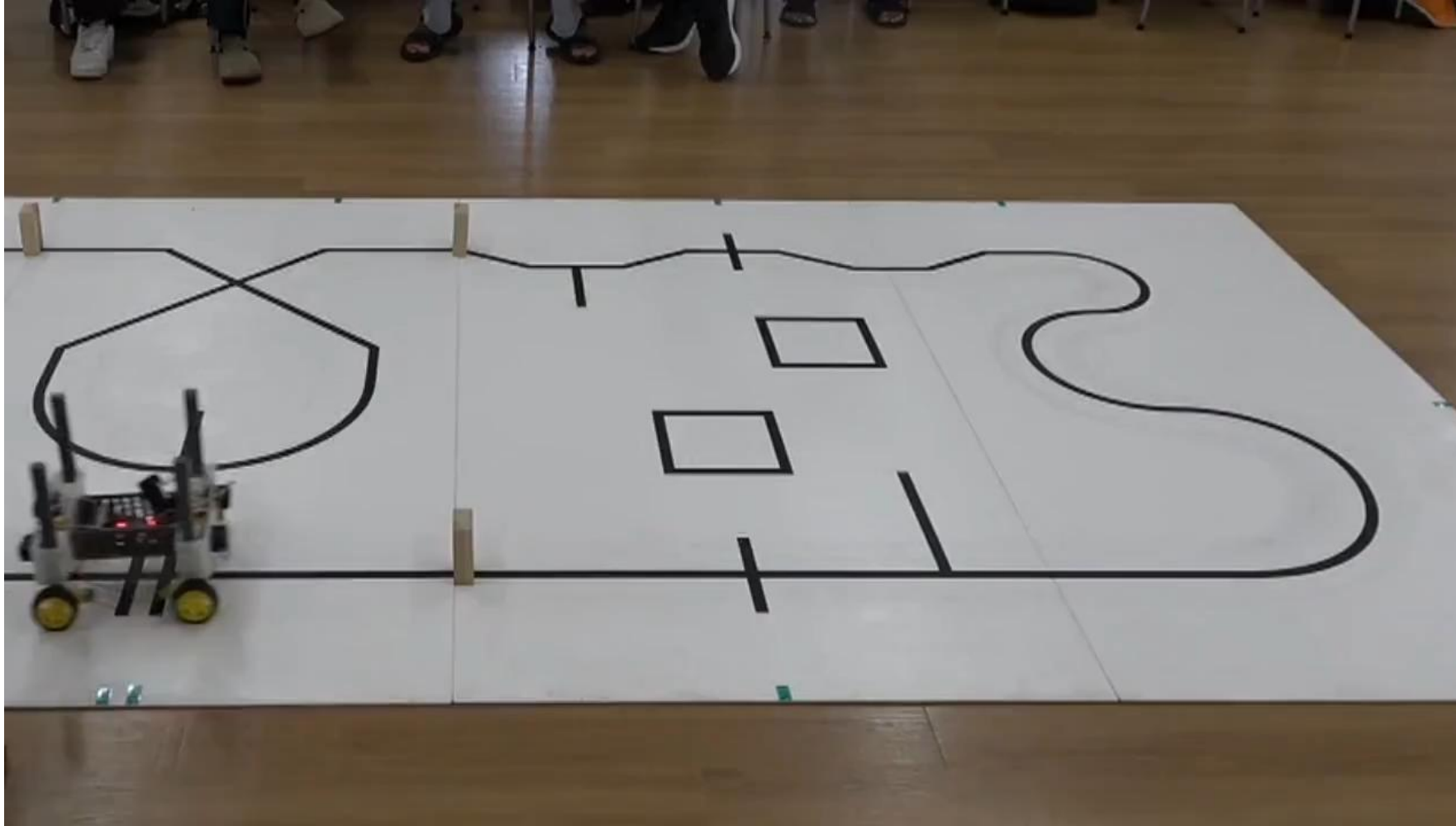


Other points than records (originality, uses of advanced technologies, etc.) are also considered in grading



[Link to the movie](#)

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[Link to the movie](#)

Example of contents in robot design plan

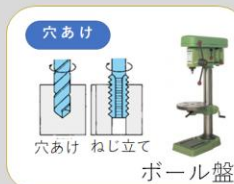
- Concept (or reasons, features, etc.) of design
- Overview of robot (layout of parts, size, etc.)
- Mechanisms
- Machining and assembling methods
(3D printers and machine tools in Machine workshop can be used if there is a good reason)
- Control algorithm
- Electric circuit diagrams
- Parts list and approximate costs

Due to the limitation of presentation, present selected topics and leave the details in the appendix.

Tools in Science plaza & machine workshop

サイエンスプラザ・機械工房 設置機器類

除去加工 Milling machine



付加加工



塑性加工



積層造形



3D printers (3 yen/g for filament)

Approval on E-portal is required, before you use.

Welding machine

Bending machine

Tools to cut materials

Band saw (K-100)

バンドソー



- Maker's page ([EN](#), [JP](#))
- Manual ([JP](#))

Hack saw (K-100)

弓のこ



Hand nibbler (TZ-20)

ハンドニブラ



- Maker's page([JP](#))
- [Movie](#)

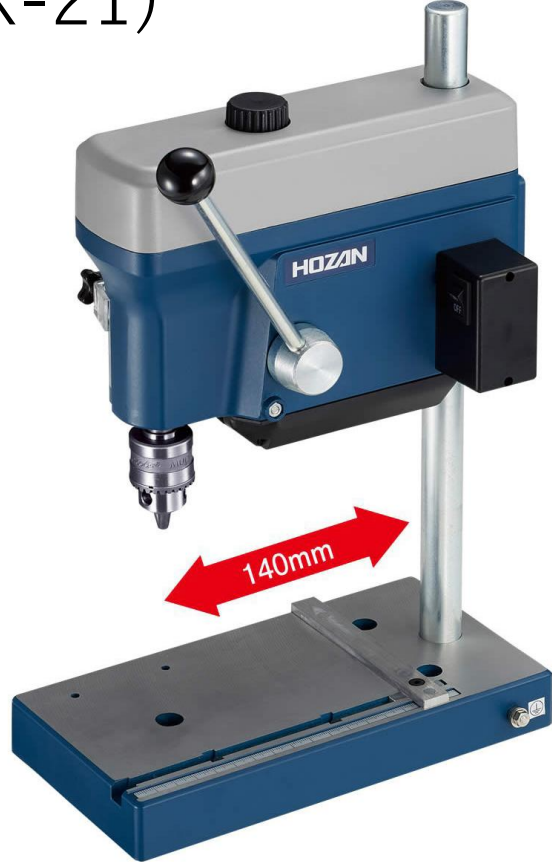
Plastic cutter(PC-L)



Tools to drill and enlarge holes

Desk drill press
(K-21)

卓上ボール盤



- Maker's page ([EN](#), [JP](#))
- Manual ([EN](#), [JP](#))

Pin vise

ピンバイス



<https://komono.me/138455>

Reamer リーマ

TR-02: diameter 4-16 mm

K-444: diameter 4~32mm



Other tools

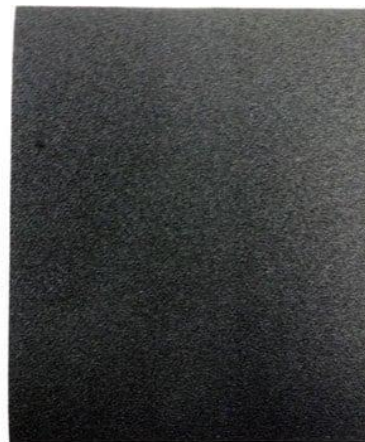
Bending & cutting machine (MSB-8)

シャーベンダー



- Maker's page ([JP](#))

File ヤスリ



Abrasive cloth sheets

研磨布シート

Hot glue gun

グルーガン



Double-sided tape

両面テープ

Examples of shops

Electronic parts, sensors, motors

- 秋月電子通商: <https://akizukidenshi.com/catalog/>
- 千石電商: <https://www.sengoku.co.jp/>
- 共立エレクトロニクス: <https://eleshop.jp/shop/default.aspx>
- スイッチサイエンス: <https://www.switch-science.com/>
- ストロベリーリナックス: <https://strawberry-linux.com/>
- Mouser (EN/JP): <https://www.mouser.jp/>
- Digi-Key (EN/JP): <https://www.digikey.jp/>
- Marutsu (JP) <https://www.marutsu.co.jp/>

Robot shop

- RobotShop (EN/JP): <https://www.robotshop.com/>
- Vstone robot shop: <https://www.vstone.co.jp/robotshop/>
- RT Robot shop: <https://www.rt-shop.jp/>

Mechanical parts, materials, etc.

- タミヤ: <https://www.tamiya.com/japan/index.html>
- モノタロウ: <https://www.monotaro.com/>
- ミスミ: <https://jp.misumi-ec.com/>
- コーナン: <https://www.kohnan-eshop.com/shop/>
- Amazon: <https://www.amazon.co.jp/>
- オリジナルマインド: <https://www.originalmind.co.jp/>
- RS コンポーネンツ: <https://jp.rs-online.com/web/>
- アクリルサンデー: <https://www.acrysunday.co.jp/>
- アルミプラス: <http://www.al-plus.jp/>

Progress report (each team): week 4

- Explain today's progress about example robot and robot design plan using slides (upload in your Team's folder)
- If you explain today's progress to me (Fukushima) and I determine that it is sufficient, you can finish today's class
- Do not leave early on your own by leaving the report to other team members
- Points will be deducted if you fail to report

Announcement

- “mbed_app.json” has been updated
 - By adding this file to your project folder, the “build” becomes faster
 - The file is in “misc” folder in Teams
- Please clean the desk and the floor around it, before you leave
 - Two vacuum cleaners in the room
 - Mini Broom & Dustpan Set at each desk

Mini Broom & Dustpan Set

Trash cans are at the back of the room

